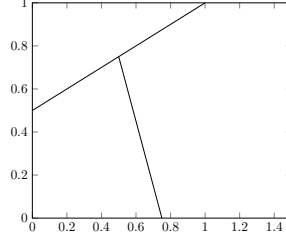


1 Mantle transport

1.1 Phase behavior

Three phases in total with 2 solid phases:

- Phase 1 can melt into fluid.
- Phase 2 can also melt into fluid.
- Different substances in fluid can react with each other.



Set up equations for component 1, which exhausts phase 1. Similarly, phase 2 is all component 2. Denote $c_{i,j}$ as component i in phase j.

Thus, the following equations can be derived:

Phase 1:

$$c_{1,1} = \rho_1 = \frac{M_{1,1}}{V_1} \quad c_{2,1} = 0 = M_{2,1} \quad (1)$$

Note: $\rho_1 = \rho_1(p_1)$ only.

Phase 2:

$$c_{1,2} = 0 = M_{1,2} \quad c_{2,2} = \rho_2 = \frac{M_{2,2}}{V_2} \quad (2)$$

Note: $\rho_2 = \rho_2(p_2)$ only.

Phase fluid:

$$c_f = c_{1,f} = \frac{M_{1,f}}{V_f}, \quad c_{2,f} = \frac{M_{2,f}}{V_f} \quad (3)$$

Additional relations and negligible diffusion:

$$\phi_f V_f \sim (\phi_1 + \phi_2) V_s, \quad \phi_f D \nabla c_f \sim 0 \quad (4)$$

Transport variable $\bar{c}$ is defined as following:

$$\begin{aligned} \bar{c} &= \phi_f c_{1,f} + \phi_1 c_{1,1} + \phi_2 c_{1,2} \\ &= \phi_f c_f + \phi_1 c_s \end{aligned} \quad (5)$$

Note

$$\phi_f = \frac{V_f}{V}, \quad \phi_1 = \frac{V_1}{V}, \quad \phi_2 = \frac{V_2}{V} \quad (6)$$

$$\phi_s = \phi_1 + \phi_2 \quad (7)$$

$$\phi_f + \phi_1 + \phi_2 = \phi_f + \phi_s = 1 \quad (8)$$

Then

$$\begin{aligned} \bar{c} &= \phi_f c_f + \phi_1 c_s = \frac{V_f}{V} \frac{M_{1,f}}{V_f} + \frac{V_1}{V} \frac{M_{1,1}}{V_1} \\ &= (M_{1,f} + M_{1,1})/V \\ &= M_1/V \end{aligned} \quad (9)$$

The transport equation is now converted into:

$$\bar{c}_t + \nabla \cdot (\bar{c}\bar{v} - \phi_f D\nabla c_f) = 0 \quad (10)$$

Two possible approaches:

(1) Linearize phase behavior:

$$\phi_f c_f = \kappa(\bar{H}, \bar{c})\bar{c} \approx \kappa\bar{c} \quad (11)$$

(2) Operator split by phase:

$$\begin{cases} (\phi_f c_f)_t + \nabla \cdot (\phi_f c_f v_f - \phi_f D\nabla c_f) = \Gamma \approx 0 \\ (\phi_1 c_s)_t + \nabla \cdot (\phi_1 c_s v_s) = -\Gamma \approx 0 \end{cases} \quad (12)$$

Flash calculation to approx Γ .

We pursue first approach (Eqn. 11). Note:

$$0 \leq \kappa \leq 1 \quad (13)$$

Since,

$$\phi_f c_f = \kappa\bar{c} = \kappa(\phi_f c_f + \phi_1 c_s)$$

moreover,

$$\bar{c} = \phi_f c_f + \phi_1 c_s$$

$\Rightarrow$

$$\phi_1 c_s = \bar{c} - \phi_f c_f = (1 - \kappa)\bar{c} \quad (14)$$

now we have

$$\begin{aligned} \bar{c}\bar{v} &= \phi_f c_f v_f + \phi_1 c_s v_s + \phi_2 \cdot 0 v_s \\ &= (\kappa v_f + (1 - \kappa)v_s)\bar{c} \end{aligned}$$

$\Rightarrow$

$$\bar{c}\bar{v} = v_e \bar{c} \quad (15)$$

$$v_e = K v_f + (1 - K)v_s \quad (16)$$

So previous equation becomes:

$$\bar{c}_t + \nabla \cdot (v_e \bar{c} - \phi_f D\nabla c_f) = 0 \quad (17)$$

Also

$$\begin{aligned} \phi_f D\nabla c_f &= \phi_f D\nabla \left(\frac{\kappa\bar{c}}{\phi_f} \right) \\ &= \kappa D\nabla \bar{c} + \bar{c} D(\nabla \kappa - \kappa \nabla \ln(\phi)) \end{aligned} \quad (18)$$

Thus transport equation (17) becomes:

$$\bar{c}_t + \nabla \cdot (v_e^* \bar{c} - \kappa D\nabla \bar{c}) = 0 \quad (19)$$

where

$$\begin{aligned} v_e^* &= v_e - D(\nabla K - K \nabla \ln(\phi)) \\ &= \kappa v_f + (1 - K)v_s - D(\nabla \kappa - \kappa \nabla \ln(\phi)) \end{aligned} \quad (20)$$

Then phase calculation for new c_f, c_s, ϕ_f, ϕ_1 .

Questions:

- 1. Is approach (11) better than (12)?
- 2. Can we approximate the nonlinearity in $\kappa(\bar{c})$ to solve (19) more accurately?
- 3. Can we get $\kappa(\bar{c})$ itself? Then solve (19) without needing flash calculation.

1.2 Thermodynamics

Let

$$E = \overline{\rho e} = \text{internal energy/volume} \quad (21)$$

Thus the energy conservation equation is defined as:

$$\frac{\partial}{\partial t} \overline{\rho e} + \nabla \cdot (\overline{\rho e v} - \overline{K} \nabla T) = 0 \quad (22)$$

Now suppose that

$$\begin{aligned} \phi_f \rho_f e_f &= \lambda(\overline{H}, \overline{C}) \overline{\rho e} = \lambda \overline{\rho e} = \lambda E \\ &= \lambda(\phi_f \rho_f e_f + \phi_1 \rho_1 e_1 + \phi_2 \rho_2 e_2) \end{aligned} \quad (23)$$

where

$$\phi_1 \rho_1 e_1 + \phi_2 \rho_2 e_2 = (1 - \lambda) \overline{\rho e} \quad (24)$$

Thus the resulting energy conservation law is:

$$E_t + \nabla \cdot (v_e^E E - \overline{K} \nabla T) = 0 \quad (25)$$

which is derived similarly with what we have done above. where average velocity is defined as:

$$v_e^E = \lambda v_f + (1 - \lambda) v_s \quad (26)$$

Now

$$\overline{K} = \phi_f K_f + \phi_1 K_1 + \phi_2 K_2 \quad (27)$$

where

$$\nabla T(C, E) = \frac{\partial T}{\partial C} \nabla C + \frac{\partial T}{\partial E} \nabla E \quad (28)$$

At each time step, temperature distribution will be obtained directly from Marc's code, as well as the corresponding spacial gradient of temperature. The convergence of this method will be researched.

1.3 Darcy-Stokes system

Quotes: The model transitions dynamically in time from a nonporous single phase Stokes to a two-phase porous medium. The Darcy part will mathematically degenerate in regions where the porosity is zero.

The resulting stokes system is described as following:

$$\begin{cases} \mathbf{u} + \frac{k_0 \phi^{2+2\Theta}}{\mu_f} \nabla q_f = 0 \\ \mu_s \nabla \cdot \mathbf{u} + \frac{\phi}{1-\phi} (q_f - q) = 0 \\ \nabla q - \nabla \cdot \hat{\boldsymbol{\sigma}}(\mathbf{v}_s) = -(1-\phi) \rho_r \mathbf{g} \\ \mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi}{1-\phi} (q_f - 1) = 0 \end{cases}$$

where pressure potentials are defined as:

$$q_f = p_f - \rho_f g z$$

$$q_s = p_s - \rho_f g z$$

and the mixture potential is therefore:

$$p_f - p_s = q_f - q_s = \frac{1}{1-\phi} (q_f - q)$$

the compatibility condition is given as:

$$\int_{\partial\Omega} (g_r + \mathbf{g}_s \cdot \boldsymbol{\nu}) ds = 0$$

The scaled formulation is also defined:

$$\tilde{\mathbf{v}}_r = \phi^{-1-\Theta} \mathbf{u}$$

$$\tilde{q}_f = \phi^{1/2} q_f$$

Then the problem is reformulated as :

$$\begin{cases} \tilde{\mathbf{v}}_r + \frac{k_0 \phi^{1+\Theta}}{\mu_f} \nabla (\phi^{1/2} q_f) = 0 \\ \mu_s \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi} (\tilde{q}_f - \phi^{1/2} q) = 0 \\ \nabla q - \nabla \cdot \hat{\boldsymbol{\sigma}}(\mathbf{v}_s) = -(1-\phi) \rho_r \mathbf{g} \\ \mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi^{1/2}}{1-\phi} (\tilde{q}_f - \phi^{1/2} q) = 0 \end{cases}$$

Implmentation of the methods on rectangular meshes:

$$\begin{pmatrix} A & -B_\phi & 0 & 0 \\ B_\phi^T & C_{f,s,\phi} & 0 & -C_{f,s,\phi} \\ 0 & 0 & D_\phi & -B \\ 0 & -C_{f,s,\phi}^T & B^T & C_{s,\phi} \end{pmatrix} \begin{pmatrix} \tilde{\mathbf{v}}_r \\ \tilde{q}_f \\ \mathbf{v}_s \\ q \end{pmatrix} = \begin{pmatrix} \alpha_\phi \\ 0 \\ b_\phi \\ 0 \end{pmatrix}$$

Using Schur complement, the remaining Stokes-like systme with two pressure potentials are solved as:

$$\begin{pmatrix} B^T A^{-1} B_\phi + C_{f,s,\phi} & -C_{f,s,\phi} \\ -C_{f,s,\phi} & B^T D^{-1} B_\phi + C_{f,s,\phi} \end{pmatrix} \begin{pmatrix} \tilde{q}_f \\ q \end{pmatrix} = \begin{pmatrix} -B_\phi^T A^{-1} \alpha_\phi \\ -B^T D^{-1} b_\phi \end{pmatrix}$$

1.4 Coupled Darcy-Stokes, transport and thermal system

The following coupled systems will be dealt with:

Darcy-Stokes system:

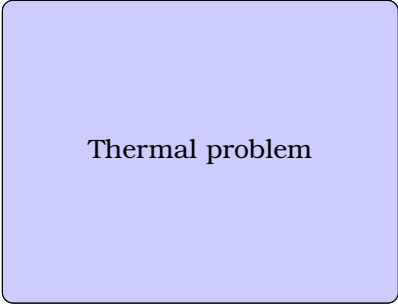
$$\left\{ \begin{array}{l} \tilde{\mathbf{v}}_r + \frac{k_0 \phi^{1+\Theta}}{\mu_f} \nabla(\phi^{1/2} q_f) = 0 \\ \mu_s \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi} (\tilde{q}_f - \phi^{1/2} q) = 0 \\ \nabla q - \nabla \cdot \hat{\boldsymbol{\sigma}}(\mathbf{v}_s) = -(1-\phi) \rho_r \mathbf{g} \\ \mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi^{1/2}}{1-\phi} (\tilde{q}_f - \phi^{1/2} q) = 0 \end{array} \right.$$

Transport and thermal system:

$$\left\{ \begin{array}{l} \bar{c}_t + \nabla \cdot (v_e^* \bar{c} - \kappa D \nabla \bar{c}) = 0 \\ E_t + \nabla \cdot (v_e^E E - \bar{K} \nabla T) = 0 \end{array} \right.$$

1.5 Spatial and time discretization

The general procedure of solving coupled stokes and transport system will be shown as follows:



Thermal problem